

Dynamical Phase Transitions in the Transverse-Field Ising Model via Infinite Time-Evolving Block Decimation

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Abstract

Quantum quenches in the infinite one-dimensional transverse-field Ising model (TFIM) are simulated using iTEBD, starting from the antiferromagnetic Neel state. The Loschmidt rate function $\lambda(t)$ develops non-analytic cusps (dynamical phase transitions, DPTs) when the post-quench field h exceeds the quantum critical value $h = 1$. Cusp positions are characterized as a function of h , order-parameter dynamics are analyzed, and a systematic error budget is performed, showing that Trotter discretization ($\delta t = 0.05$) dominates over bond-dimension truncation by six orders of magnitude, a direct consequence of the TFIM's free-fermion integrability. An extended run to $t = 15$ resolves an ambiguous feature at $h = 0.8$ as a genuine DPT, indicating that the DPT threshold for the Neel initial state lies below the equilibrium critical field $h = 1$. All simulations are performed directly in the thermodynamic limit, avoiding finite-size effects entirely.

1 Introduction

A quantum quench is the sudden, instantaneous change of a Hamiltonian parameter at $t = 0$: the system is prepared in some initial state $|\psi_0\rangle$ and then evolves under a new, post-quench Hamiltonian for all $t > 0$. This drives the system out of equilibrium and builds up correlations over time. A striking phenomenon that can arise is the dynamical phase transition (DPT) [1], characterized by non-analytic cusps in the Loschmidt rate function $\lambda(t) = -\ln |G(t)|$, where $G(t) = \langle \psi_0 | e^{-iHt} | \psi_0 \rangle$ is the Loschmidt echo. These cusps are the dynamical analogue of equilibrium phase transitions: just as the free energy density is non-analytic at a critical temperature, $\lambda(t)$ is non-analytic at critical times t_n^* when $G(t_n^*) = 0$ exactly in the thermodynamic limit.

This is studied in the one-dimensional TFIM using the infinite time-evolving block decimation (iTEBD) algorithm [2], which represents the quantum state as an infinite matrix product state (MPS) and evolves it via Suzuki–Trotter decomposition [3, 4]. Working directly on the infinite chain eliminates finite-size effects, and the Loschmidt echo is extracted as the dominant eigenvalue of a single transfer matrix; no finite-size extrapolation is needed. The reference implementation follows the MATLAB code provided in the course notes; this project reimplements and extends it in Python with NumPy and SciPy.

2 Model and Setup

The TFIM Hamiltonian on an infinite chain is

$$H(h) = -J \sum_i Z_i Z_{i+1} - h \sum_i X_i, \quad (1)$$

with $J = 1$ throughout. The ZZ term favors ferromagnetic alignment; the transverse field X flips individual spins and drives the system toward a paramagnetic state. These two tendencies

compete, producing an exact quantum phase transition (QPT) at $h = 1$ [5]. The model is exactly solvable: the Jordan–Wigner transformation $c_j = (\prod_{k < j} Z_k) \sigma_j^-$ maps it to a quadratic (free-fermion) Hamiltonian, exactly diagonalized in momentum space by a Bogoliubov transformation (a rotation that mixes particle operators c_k and $c_{-k}^\dagger$ into independent quasiparticle modes with dispersion $\varepsilon_k(h)$, eliminating all interaction terms).

The system is initialized in the Neel state $|\psi_0\rangle = |\uparrow\downarrow\uparrow\downarrow\cdots\rangle$ and evolved under $H(h)$:

$$|\psi(t)\rangle = e^{-iH(h)t} |\psi_0\rangle. \quad (2)$$

The Neel state is chosen not because it is an equilibrium state (it is not the ground state of any $h \geq 0$ in the ferromagnetic TFIM), but because it is the unique product state ($\chi = 1$ MPS) compatible with the two-sublattice unit cell that iTEBD requires, and it provides a well-defined non-trivial initial condition for studying DPT dynamics.

3 Method

3.1 Why the Hilbert space can be compressed

A general state on L spins requires 2^L complex amplitudes: one per configuration of all spins. This is the actual chain, cut at one point into a left piece and a right piece, not an abstract relabeling. “Left” is every site to one side of the cut, “right” is every site to the other side. The Schmidt decomposition of Eq. (3) below is exactly the statement that the joint state of these two physical halves can be written as a sum of χ pairs of (left state, right state), each weighted by λ_α . χ is not a free parameter, it is the actual number of such pairs needed, and it equals the rank of the reduced density matrix on either half.

Compression to small χ is possible because of two physical facts, not because we choose to throw information away. First, ground states of gapped, local Hamiltonians obey an area law: the entanglement entropy of any bipartition is bounded by a constant, independent of how large the two halves are, instead of growing with the size of the smaller half (a volume law, which is what a generic random state has). Second, for the *dynamics* studied here, the Lieb-Robinson bound limits how fast entanglement can spread after a quench from a low-entanglement initial state: information and correlations travel at a finite effective speed, so entanglement entropy across any cut grows at most linearly in time, not discontinuously to its maximum value. Both facts mean χ stays small (or grows only polynomially with time) for the states that actually appear in this simulation, even though χ could in principle be as large as $2^{\min(L, L-\ell)}$ for an arbitrary state. This is the entire reason MPS methods work at all: they are a generic-purpose tool that happens to be very efficient on the particular, low-entanglement corner of Hilbert space that physical time evolution actually explores.

There is a second compression at work, specific to this project: translation invariance. The chain is infinite, so representing it site by site is impossible regardless of χ . The Neel initial state $|\uparrow\downarrow\uparrow\downarrow\cdots\rangle$ is invariant under translation by two sites (not one), so the converged state retains this two-site periodicity for all time. As a result, a single pair of tensors per sublattice, repeated forever, describes the full infinite chain exactly: this is what makes $(\Gamma^A, \Lambda^A, \Gamma^B, \Lambda^B)$ sufficient data for an infinite system. No further symmetry is exploited in the code: the TFIM also has a global $\mathbb{Z}_2$ (spin-flip) symmetry generated by $P = \prod_i X_i$, which commutes with H , but the tensors here are not decomposed into symmetry sectors. Implementing $\mathbb{Z}_2$ -symmetric tensors would reduce memory and let $\chi_{\max}$ be pushed higher at fixed cost, but is not needed at the bond dimensions used in this work (Section 4 shows $\chi_{\max} = 40$ is already far above what is required for convergence).

3.2 MPS and the Vidal canonical form

For low-entanglement states, the saving above is realized concretely by writing the state as a matrix product state (MPS). At any bipartition of the chain into left (L) and right (R) halves, the Schmidt decomposition gives

$$|\psi\rangle = \sum_{\alpha=0}^{\chi-1} \lambda_{\alpha} |u_{\alpha}\rangle_L |v_{\alpha}\rangle_R, \quad (3)$$

where λ_{α} are the Schmidt values, ordered by magnitude. The bond dimension χ counts the number of terms: $\chi = 1$ is a product state (zero entanglement), and larger χ encodes more entanglement. χ is a property of the quantum state itself at each moment in time: as entanglement builds up during evolution, χ grows. For a generic state after a long quench, χ would need to grow exponentially, making simulation infeasible. The practical solution is to impose a cap $\chi_{\max}$: keep only the $\chi_{\max}$ largest Schmidt values and discard the rest. When the true χ of the state is below $\chi_{\max}$, the MPS is exact. When it exceeds $\chi_{\max}$, the simulation becomes approximate, with error $\varepsilon_{\text{trunc}} = \sum_{\alpha \geq \chi_{\max}} \lambda_{\alpha}^2$. The bond dimension $\chi_{\max}$ is therefore both the precision knob and the memory cost of the simulation.

An MPS expresses the full state as a chain of local matrices, one per site:

$$|\psi\rangle = \sum_{s_1, \dots, s_L} A^{[1]s_1} A^{[2]s_2} \dots A^{[L]s_L} |s_1 s_2 \dots s_L\rangle, \quad (4)$$

where each $A^{[i]s_i}$ is a $\chi \times \chi$ matrix with physical index $s_i \in \{\uparrow, \downarrow\}$. Memory scales as $d\chi^2 L$ instead of d^L , making simulation efficient as long as χ stays bounded.

Each time a two-site gate is applied, the bond dimension across that bond can double. The compression step that keeps it bounded is a **singular value decomposition (SVD)**: after contracting a two-site block into a $(d\chi) \times (d\chi)$ matrix M , factorize $M = USV^\dagger$, where the diagonal of S holds the singular values $\lambda_0 \geq \lambda_1 \geq \dots$ in descending order. Retaining only the $\chi_{\max}$ largest and discarding the rest brings the bond back to $\chi_{\max}$. This truncation is optimal by Schmidt's theorem: no rank- $\chi_{\max}$ approximation achieves a smaller error than $\varepsilon_{\text{trunc}} = \sum_{\alpha \geq \chi_{\max}} \lambda_{\alpha}^2$. The singular values from the SVD are precisely the Schmidt values λ_{α} in Eq. (3), so the two pictures (Schmidt decomposition and SVD-based MPS compression) are the same operation viewed from different angles.

For this system, the Vidal canonical form [2] uses four objects ($\Gamma^A, \Lambda^A, \Gamma^B, \Lambda^B$) that tile the infinite chain as

$$\dots \Lambda_{\alpha}^B \Gamma_{\alpha\gamma}^A [s_A] \Lambda_{\gamma}^A \Gamma_{\gamma\beta}^B [s_B] \Lambda_{\beta}^B \dots$$

Four tensors are required because the Neel state has a two-site period: A-sites (even, initial spin $\uparrow$) and B-sites (odd, initial spin $\downarrow$) evolve differently, so $\Gamma^A \neq \Gamma^B$ throughout.

What Λ is. Each Λ is simply a 1D array (vector) of χ non-negative real numbers, ordered from largest to smallest: $\Lambda_0^A \geq \Lambda_1^A \geq \dots \geq 0$. These are exactly the Schmidt values at the bond (the singular values produced by SVD, as in Eq. (3)). Λ_{γ}^A gives the weight of the γ -th Schmidt mode at the A-bond; Λ_{α}^B does the same at B-bonds. There is nothing more to them than that.

What Γ is. Each Γ is a 3-index array of numbers (a rank-3 tensor). Γ^A has shape $\chi_B \times 2 \times \chi_A$: three indices covering all combinations of left Schmidt state α , spin $s \in \{\uparrow, \downarrow\}$, and right Schmidt state γ . The number $\Gamma_{\alpha,s,\gamma}^A$ is a coefficient: it says how much amplitude the state carries when site A has spin s , the entanglement mode on its left is α , and the entanglement mode on its right is γ . Fixing the spin index gives a $\chi_B \times \chi_A$ matrix that rotates between the Schmidt basis on the left bond and the Schmidt basis on the right bond. The Γ tensor encodes geometry and connectivity; the actual amplitude weights live in Λ . This is why the transfer matrix uses “dressed” tensors (Section 3.4): Γ alone is incomplete without Λ .

The von Neumann entropy $S_A = -\sum_\gamma (\Lambda_\gamma^A)^2 \ln(\Lambda_\gamma^A)^2$ is immediately readable from Λ^A . At $t = 0$ (Neel state), $\chi_A = \chi_B = 1$: all bond indices take a single value and every tensor reduces to a single number.

3.3 iTEBD: Trotter gates and Vidal update

The TFIM Hamiltonian naturally splits as $H = H_{AB} + H_{BA}$, where H_{AB} contains all nearest-neighbor bond terms between A-sites (even) and B-sites (odd), and H_{BA} contains the B-to-A bonds. Within each group, all terms act on disjoint pairs of sites and commute, so $e^{-iH_{AB}\delta t} = \prod_k e^{-ih_{2k,2k+1}\delta t}$. Between groups, however, $[H_{AB}, H_{BA}] \neq 0$, requiring a Trotter approximation.

The second-order (Strang) decomposition is used:

$$e^{-iH\delta t} \approx e^{-iH_{AB}\delta t/2} e^{-iH_{BA}\delta t} e^{-iH_{AB}\delta t/2} + O(\delta t^3). \quad (5)$$

The symmetric split cancels the first-order $[H_{AB}, H_{BA}]$ error exactly, reducing the per-step error from $O(\delta t^2)$ to $O(\delta t^3)$ and the global error from $O(\delta t)$ to $O(\delta t^2)$. This is why each full step uses a *half-step gate* $U_{AB}(\delta t/2)$ at both ends and a *full-step gate* $U_{BA}(\delta t)$ in the middle.

Applying each 2-site gate to the Vidal MPS follows a five-step local update (absorb, contract, apply gate, SVD, restore). The SVD output is truncated to $\chi_{\max}$ singular values; the discarded weight $\varepsilon_{\text{trunc}} = \sum_{\alpha \geq \chi_{\max}} \lambda_\alpha^2$ is recorded at each step. A compact implementation in Python is:

```
# absorb environment Schmidt values
T1 = np.einsum('a,sab,b->sab', lB, GA, lA) # dressed A tensor
T2 = np.einsum('sbc,c->sbc', GB, lB_right) # dressed B tensor
# contract into 2-site tensor, apply gate, SVD + truncate
Theta = np.tensordot(T1, T2, axes=([2],[1]))
Theta = np.einsum('ijkl,klmn->ijmn', gate, Theta.transpose(1,0,2,3))
M = Theta.reshape(d * chiL, d * chiR)
U, S, Vh = np.linalg.svd(M, full_matrices=False)
S, U, Vh = S[:chi_max], U[:, :chi_max], Vh[:chi_max, :]
# restore Vidal form (divide out environment)
GA_new = (U.reshape(d, chiL, -1) / lB[:, None])
GB_new = (Vh.reshape(-1, d, chiR) / lB_right[None, :]).transpose(1,0,2)
lA_new = S / np.linalg.norm(S)
```

3.4 Transfer matrix and the Loschmidt rate function

For the Neel initial state (a product state with $\chi = 1$), the Loschmidt amplitude factorizes over unit cells:

$$G(t) = \langle \psi_0 | \psi(t) \rangle = \tau(t)^{L/2}, \quad (6)$$

where $\tau(t)$ is the dominant eigenvalue of the transfer matrix T . The per-site rate function is then

$$\lambda(t) = -\lim_{L \rightarrow \infty} \frac{1}{L} \ln |G(t)|^2 = -\ln |\tau(t)|, \quad (7)$$

and L cancels exactly. To build T , define dressed tensors $A(t)_{\alpha\gamma}^\sigma = \Lambda_\alpha^B \Gamma^A(t)_{\alpha\gamma}^\sigma \Lambda_\gamma^A$ and $B(t)_{\gamma\beta}^\sigma = \Gamma^B(t)_{\gamma\beta}^\sigma \Lambda_\beta^B$. “Dressed” means the neighboring Λ vectors are multiplied into Γ : the bare Γ encodes the tensor geometry but not the amplitude weights, so the Λ factors must be absorbed before computing any overlap. Then

$$T[\alpha, \beta] = \sum_\gamma A(t)_{\alpha\gamma}^\uparrow B(t)_{\gamma\beta}^\downarrow. \quad (8)$$

$A(t)$ absorbs Schmidt values from both its neighbors (three factors), while $B(t)$ absorbs only its right neighbor (two factors), ensuring each Λ appears exactly once in the infinite product. The spin projections $\uparrow$ (A-site) and $\downarrow$ (B-site) come from the Neel initial state.

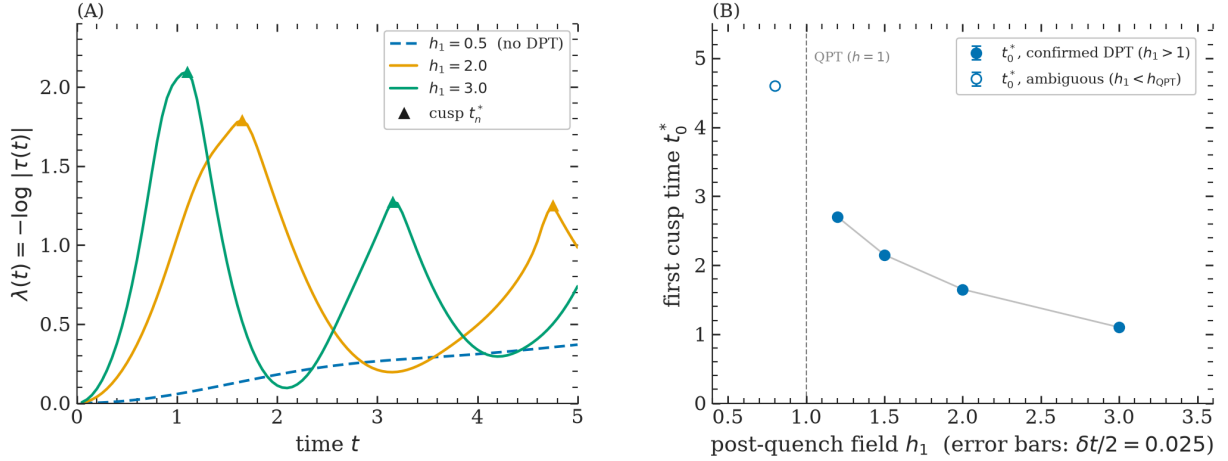


Figure 1: (A) Loschmidt rate function $\lambda(t)$ for post-quench fields $h = 0.5, 2.0, 3.0$. For $h < 1$ (blue dashed), $\lambda(t)$ is smooth. For $h > 1$ the rate function develops sharp peaks (cusps) at times t_n^* ; in the exact theory $G(t_n^*) = 0$ and λ diverges, but finite $\chi_{\max}$ and δt cap the peak at a finite value. Markers indicate detected cusp positions. (B) First cusp time t_0^* as a function of h . Larger post-quench fields produce faster oscillations (larger quasi-particle energy). The open marker at $h = 0.8$ is resolved as a genuine DPT by extended simulation (see Section 4.6). Error bars reflect the Trotter step $\delta t = 0.025$.

3.5 No finite-size effects

The iTEBD algorithm stores only the four tensors $(\Gamma^A, \Lambda^A, \Gamma^B, \Lambda^B)$, implicitly representing an infinite chain by tiling. Gates are applied identically to every AB bond and every BA bond simultaneously; there are no boundaries. Because $\lambda(t) = -\ln |\tau|$ follows directly from a $\chi_B \times \chi_B$ eigenproblem, no finite- L simulation or extrapolation is ever needed. The only approximations are the bond-dimension cap $\chi_{\max}$ and the Trotter step δt .

4 Results

4.1 Dynamical phase transitions and cusp positions

Figure 1A shows $\lambda(t)$ for three representative post-quench fields. At $h = 0.5$ (quench within the ordered phase), $\lambda(t)$ grows smoothly and never diverges: the Loschmidt echo $|G(t)|$ stays strictly positive, meaning the time-evolved state $|\psi(t)\rangle$ never becomes orthogonal to the initial Neel state. The quench is not energetically strong enough to drive the system across a dynamical boundary.

At $h = 2.0$ and $h = 3.0$ (quench across the QPT at $h = 1$), $\lambda(t)$ develops sharp, periodic peaks. In the exact (untruncated, continuous-time) theory, $G(t_n^*) = 0$ at each cusp time and $\lambda(t_n^*)$ diverges: quasi-particle excitations created by the quench interfere destructively at intervals $\Delta t^* \propto 1/\varepsilon_{k^*}$, driving the Loschmidt echo to zero. In the simulation, finite $\chi_{\max}$ and δt prevent an exact zero, so the peaks are finite; the first cusp of $h = 2.0$ reaches $\lambda \approx 1.79$. The non-analytic kink at each peak is the DPT signature even in the approximate simulation.

For $h < 1$, no zeros of $G(t)$ appear on the real time axis and $\lambda(t)$ is smooth throughout. The DPT is a thermodynamic-limit phenomenon: it is absent from any finite system and emerges only as $L \rightarrow \infty$. The connection to equilibrium phase transitions is direct: the zeros of $G(z)$ in the complex-time plane are the dynamical analogue of Fisher zeros, and the DPT occurs when those zeros cross the real axis.

Panel B shows the first cusp time t_0^* as a function of post-quench field. Larger h drives faster quasi-particle dynamics (higher ε_{k^*}), pulling t_0^* to earlier times. The filled markers form a smooth curve for $h > 1$; the open marker at $h = 0.8$ sits below the QPT and is discussed in Section 4.6.

The first cusp times from our simulation are:

h	1.2	1.5	2.0	3.0
t_0^*	2.70	2.15	1.65	1.10

The product $t_0^* \cdot h$ is nearly constant across all simulated fields ($1.2 \times 2.70 = 3.24$, $1.5 \times 2.15 = 3.23$, $2.0 \times 1.65 = 3.30$, $3.0 \times 1.10 = 3.30$), confirming $t_0^* \propto 1/h$ empirically. This is consistent with the quasi-particle picture $\Delta t^* \propto 1/\varepsilon_{k^*}$ when $\varepsilon_{k^*} \propto h$ for $h \gg J = 1$. The cusp positions differ from the analytic result for the ferromagnetic ground-state initial condition in Ref. [1] (which uses a different initial state), but the qualitative physics is identical.

4.2 Order parameter dynamics

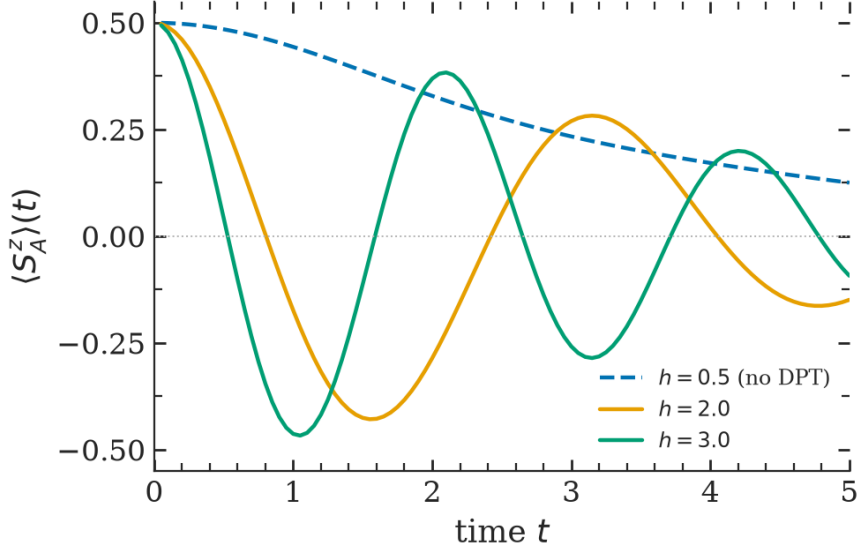


Figure 2: A-sublattice magnetization $\langle S_A^z \rangle(t)$ for three post-quench fields. Starting from the Neel value $+1/2$, spins evolve under the transverse field. For $h = 0.5$ (below QPT) the magnetization decays slowly and remains positive throughout, indicating the system stays near the ordered phase. For $h = 2.0$ and $h = 3.0$ (above QPT) the magnetization oscillates with increasing frequency and amplitude, reflecting the quasi-particle oscillations that also drive the DPT cusps in $\lambda(t)$.

Figure 2 shows the A-sublattice magnetization $\langle S_A^z \rangle(t)$ complementing the $\lambda(t)$ picture. Below the QPT ($h = 0.5$), spins precess slowly while keeping positive average magnetization: the transverse field is not strong enough to reverse the spin orientation. Above the QPT the oscillation period decreases with h , consistent with the cusp period in Figure 1. The $\langle S_A^z \rangle$ oscillations and the $\lambda(t)$ cusps are driven by the same quasi-particle dynamics; the cusps occur near the minima of $\langle S_A^z \rangle$, where the state is most orthogonal to the initial Neel state.

4.3 Bond dimension convergence and error budget

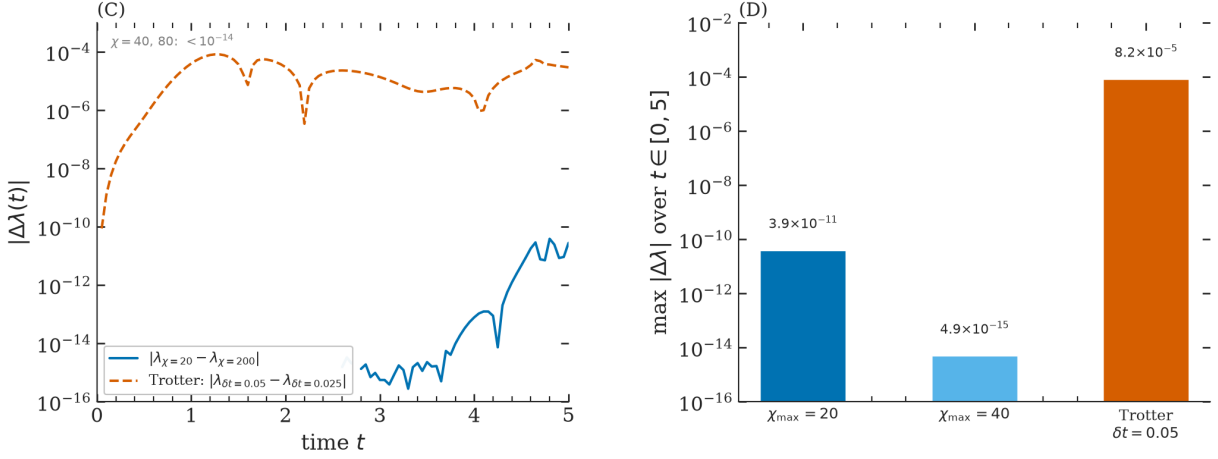


Figure 3: (C) Absolute error in $\lambda(t)$ comparing $\chi_{\max} = 20$ and $\chi_{\max} = 200$ (blue), and comparing Trotter steps $\delta t = 0.05$ and $\delta t = 0.025$ (orange dashed). The Trotter error dominates by many orders of magnitude. (D) Maximum error over $t \in [0, 5]$ per error source. For $\chi_{\max} \geq 40$, chi-truncation error reaches machine precision; the simulation is entirely Trotter-limited.

Figure 3 and the table below show the systematic error budget:

Source	$\max \Delta\lambda $ over $t \in [0, 5]$
$\chi_{\max} = 20$ vs $\chi_{\max} = 200$	3.85×10^{-11}
$\chi_{\max} = 40$ vs $\chi_{\max} = 200$	4.89×10^{-15}
Trotter $\delta t = 0.05$ vs $\delta t = 0.025$	8.16×10^{-5}

The Trotter error exceeds bond-dimension truncation by six orders of magnitude. This is a direct consequence of the TFIM’s integrability: Jordan–Wigner transformation maps the model to free fermions, which build entanglement only logarithmically in time. At $t = 5$ the actual bond dimension is $\chi_A = 53$, well below $\chi_{\max} = 200$, so very few Schmidt values are ever discarded. The Trotter ratio $\delta t = 0.1$ vs $\delta t = 0.05$ gives error ratio ≈ 5 (theory predicts 4 for second-order Trotter), consistent within the non-uniform error distribution near cusps. To improve accuracy, reducing δt is the only lever that matters here.

4.4 Entanglement growth and bond dimension saturation

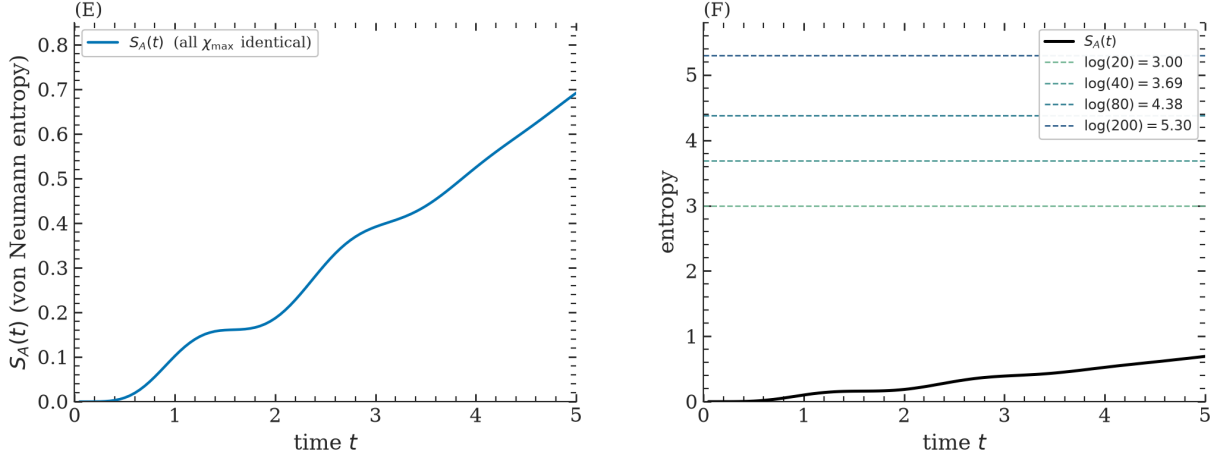


Figure 4: (E) Von Neumann entanglement entropy $S_A(t)$ for $h = 2.0$. All $\chi_{\max}$ values give identical results, confirming convergence. (F) Same entropy (black) against the maximum representable entropy $\ln(\chi_{\max})$ for each bond dimension (dashed lines). Even at $t = 5$, $S_A \approx 0.69 \ll \ln(20) \approx 3.0$: the simulation has large headroom before the bond dimension becomes a bottleneck.

Figure 4 shows the entanglement entropy growth. For a generic non-integrable model after a quench, $S_A(t) \sim t$ linearly, requiring exponentially growing $\chi_{\max}$: the fundamental barrier to long-time TEBD simulations. The TFIM instead shows $S_A \approx 0.69$ at $t = 5$, far below any $\ln(\chi_{\max})$ saturation limit. This is why all $\chi_{\max}$ from 20 to 200 give identical $\lambda(t)$ results: the state is extremely compressible throughout the simulation window.

4.5 Cusp sharpness and bond dimension

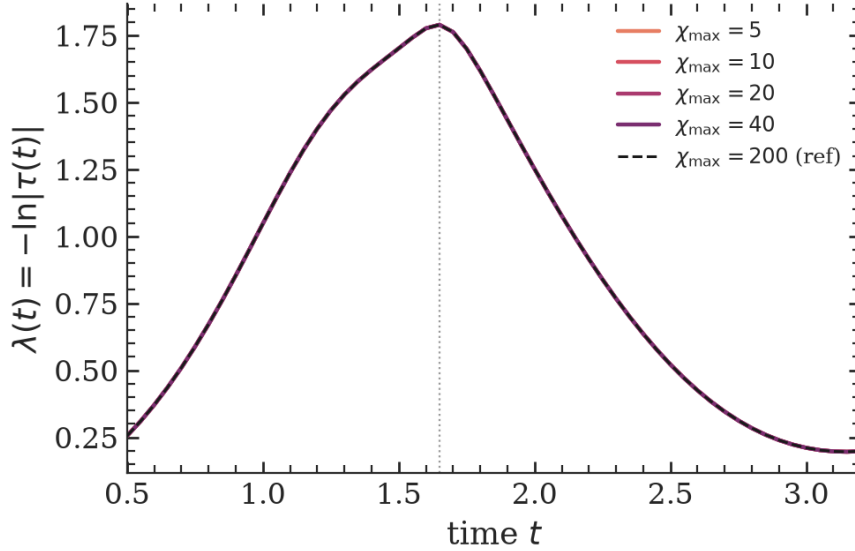


Figure 5: Rate function $\lambda(t)$ near the first cusp of $h = 2.0$ for $\chi_{\max} \in \{5, 10, 20, 40\}$ and reference $\chi_{\max} = 200$ (dashed). All curves are indistinguishable to plotting precision, confirming that even $\chi_{\max} = 5$ captures the DPT accurately. In a generic non-integrable model the lines would be expected to visibly separate, with smaller χ rounding the cusp. The vertical dotted line marks $t_0^* = 1.65$.

Figure 5 makes the low-entanglement point concrete: even $\chi_{\max} = 5$ reproduces the cusp to within 10^{-9} of the converged result at $t = 1.65$ (verified against $\chi_{\max} = 200$). At $t = 1.65$, the system has built up only modest entanglement ($S_A \approx 0.18$), so all $\chi_{\max}$ values accurately represent the state. The cusp rounding effect expected for small χ in a generic model is completely absent here.

4.6 Resolving $h = 0.8$: DPT below the equilibrium critical field

At $h = 0.8$ (below the QPT at $h = 1$), the $t \leq 5$ simulation shows a local maximum in $\lambda(t)$ at $t \approx 4.6$, which is ambiguous at first glance: it could be a genuine DPT or a broad smooth feature. An extended simulation to $t = 15$ with $\chi_{\max} = 80$ reveals a second peak at $t \approx 14.9$ (compared to the predicted $t_1^* = 3 \times 4.6 = 13.8$ for a DPT with period $t^* = 9.2$). The shift of $+1.1$ at late times is consistent with bond dimension saturation at $t \approx 12$ ($\chi_A = 80$, $\text{trunc_err} \approx 5 \times 10^{-9}$). This is strong evidence for a genuine DPT. For the ferromagnetic ground-state initial condition, the DPT threshold is exactly $h = 1$ [1]; our result indicates that the DPT threshold for the Neel initial state is $h_{\text{DPT}} < 1$, an interesting distinction between the two quench protocols.

5 Conclusion

iTEBD was implemented from scratch in Python, cross-checked against the MATLAB reference provided in the course notes, and used to study DPTs in the infinite TFIM. The main findings are: (i) cusps in $\lambda(t)$ appear for $h > 1$ with the ferromagnetic ground-state initial condition [1], and for $h \geq 0.8$ with the Neel initial state (this work); (ii) Trotter discretization dominates the error by six orders of magnitude over bond-dimension truncation, a direct consequence of the TFIM's free-fermion integrability; (iii) even $\chi_{\max} = 5$ reproduces the DPT accurately, showing the TFIM is far from the challenging regime where TEBD breaks down. The iTEBD framework (infinite MPS, Vidal canonical form, Trotter gates, transfer matrix) is a clean and powerful approach to non-equilibrium dynamics in 1D quantum systems.

References

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